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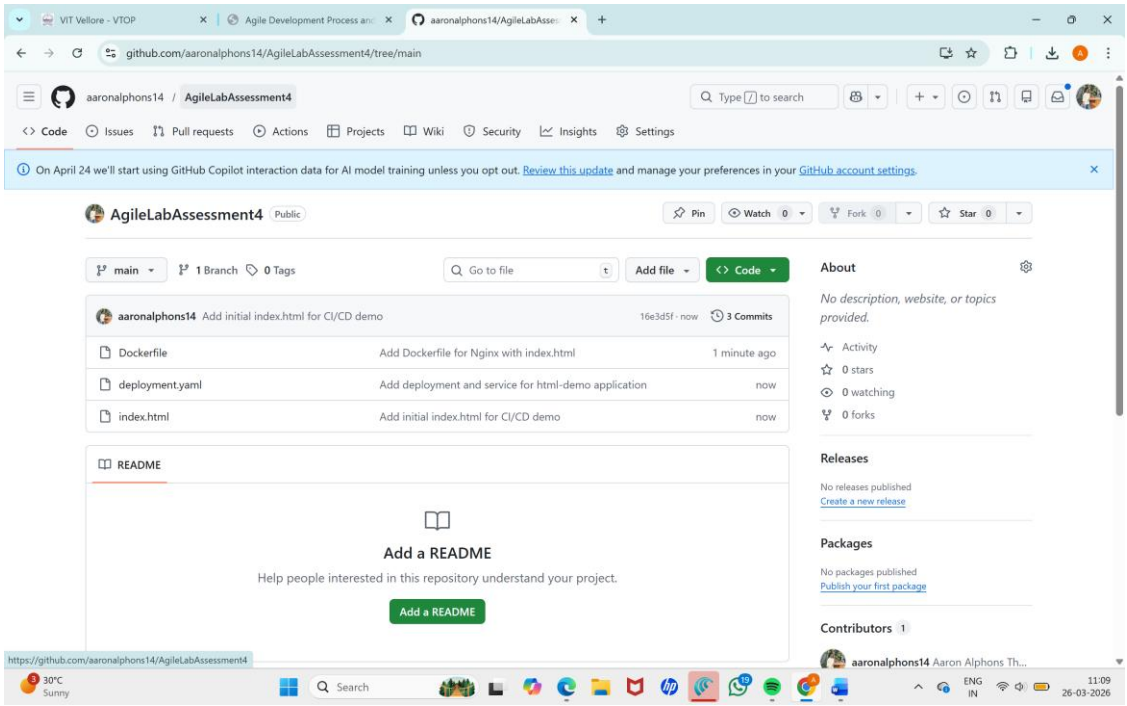
Agile and DevOps lab Assignment 4 Document

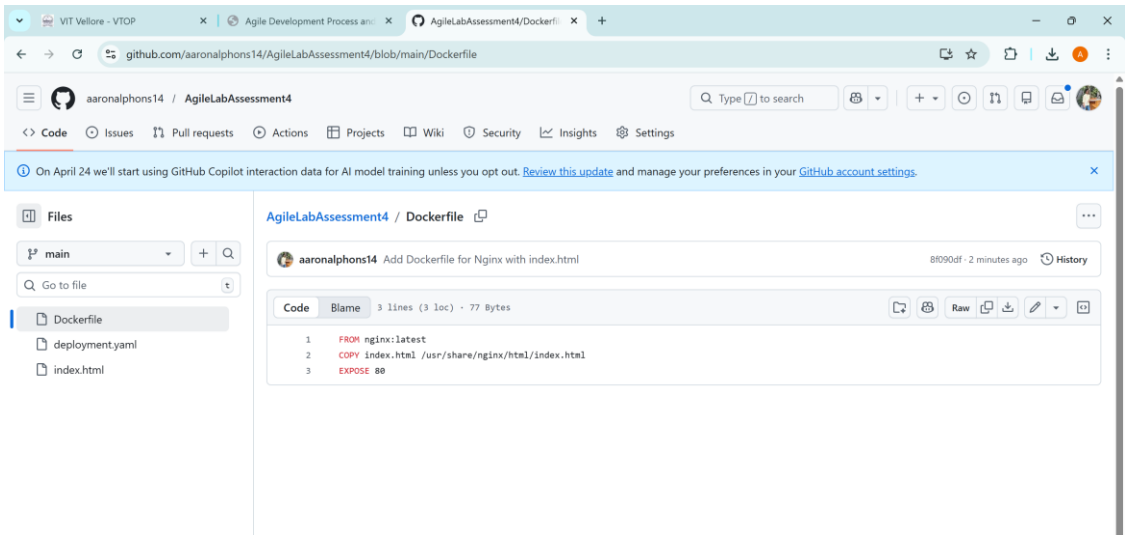
Assignment:

Create an HTML file for your profile and complete the following:

1.Create a repository in GIT and create the following files:

- **index.html**
- **Dockerfile**
- **Deployment.yaml**





github.com/aaronalphons14/AgileLabAssessment4/blob/main/deployment.yaml

aaronalphons14 / AgileLabAssessment4

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.

Files

main

Go to file

Dockerfile

deployment.yaml

index.html

AgileLabAssessment4 / deployment.yaml

aaronalphons14 Add deployment and service for html-demo application

289db3e · 1 minute ago History

Code Blame 32 lines (32 loc) · 443 Bytes

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: html-demo
spec:
 replicas: 2
 selector:
 matchLabels:
 app: html-demo
 template:
 metadata:
 labels:
 app: html-demo
 spec:
 containers:
 - name: html-demo
 image: sweetab04/html-demo:latest
 ports:
 - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
 name: html-demo-service
spec:
 type: NodePort
 selector:
 app: html-demo
 ports:
 - port: 80
 targetPort: 80
 nodePort: 30010
```

30°C Sunny Search ENG IN 11:11 26-03-2026

github.com/aaronalphons14/AgileLabAssessment4/blob/main/index.html

aaronalphons14 / AgileLabAssessment4

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.

Files

main

Go to file

Dockerfile

deployment.yaml

index.html

AgileLabAssessment4 / index.html

aaronalphons14 Add initial index.html for CI/CD demo

16e3d5f · 2 minutes ago History

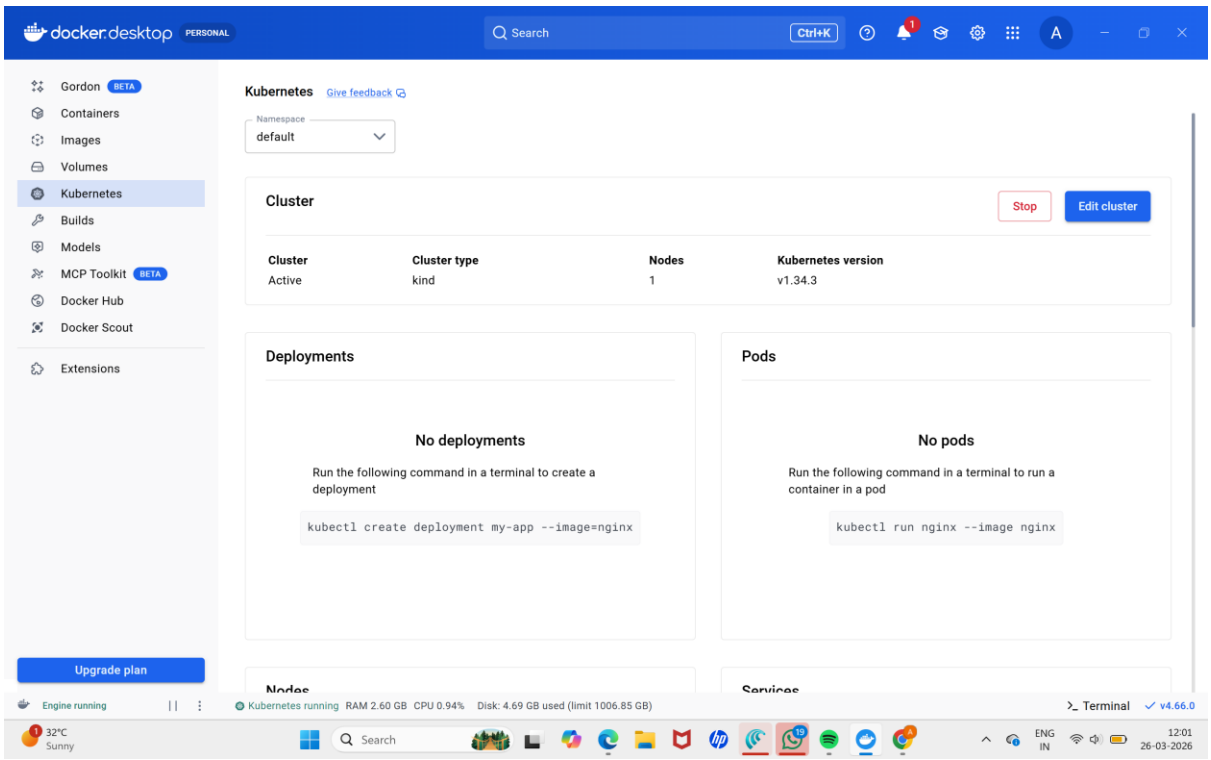
Code Blame 21 lines (21 loc) · 363 Bytes

```html
<!DOCTYPE html>
<html>
<head>
<title>CI/CD Demo</title>
<style>
body{
text-align:center;
font-family:Times New Roman;
background-color:#af0d0d;
}
h1{
color:#0ef777;
}
</style>
</head>
<body>
<h1>CI/CD Pipeline Working</h1>
<h2>Deployed using Jenkins + Docker + Kubernetes</h2>
<p>This webpage is automatically deployed using a CI/CD pipeline.</p>
</body>
</html>
```

30°C Sunny Search ENG IN 11:11 26-03-2026

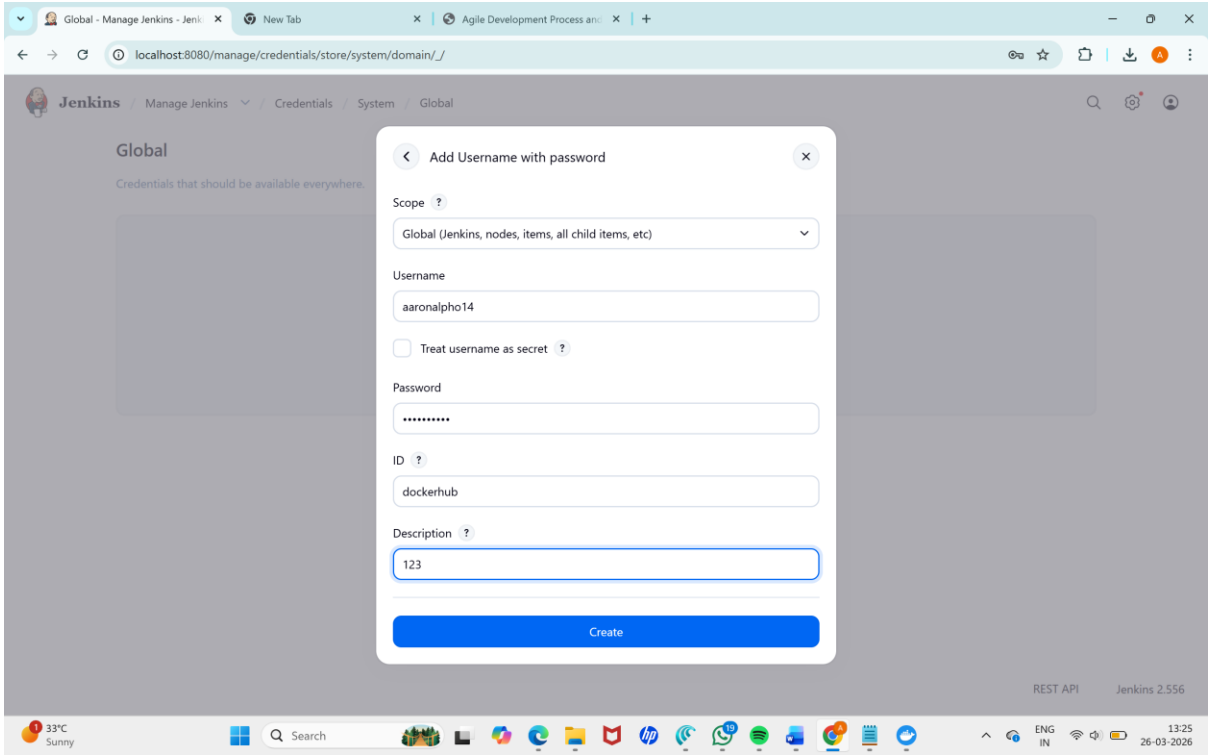
2. Enable Kubernetes

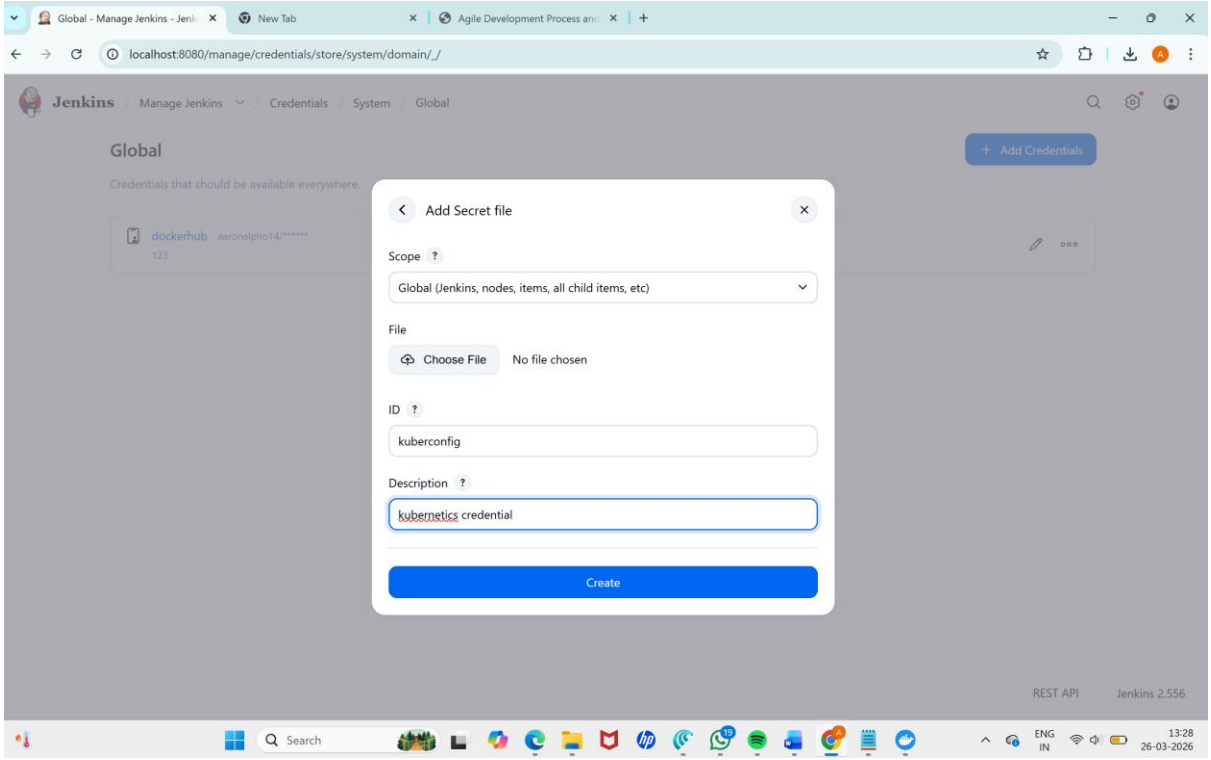
- **Open the Docker Dashboard: Launch the Docker Desktop application on your system.**
- **Navigate to Settings: Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.**
- **Select the Kubernetes Tab: In the Settings menu, find and select the Kubernetes tab.**
- **Enable Kubernetes: Check the box labeled Enable Kubernetes.**
- **Apply & Restart: Click the Apply & Restart button at the bottom of the settings window. Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time**



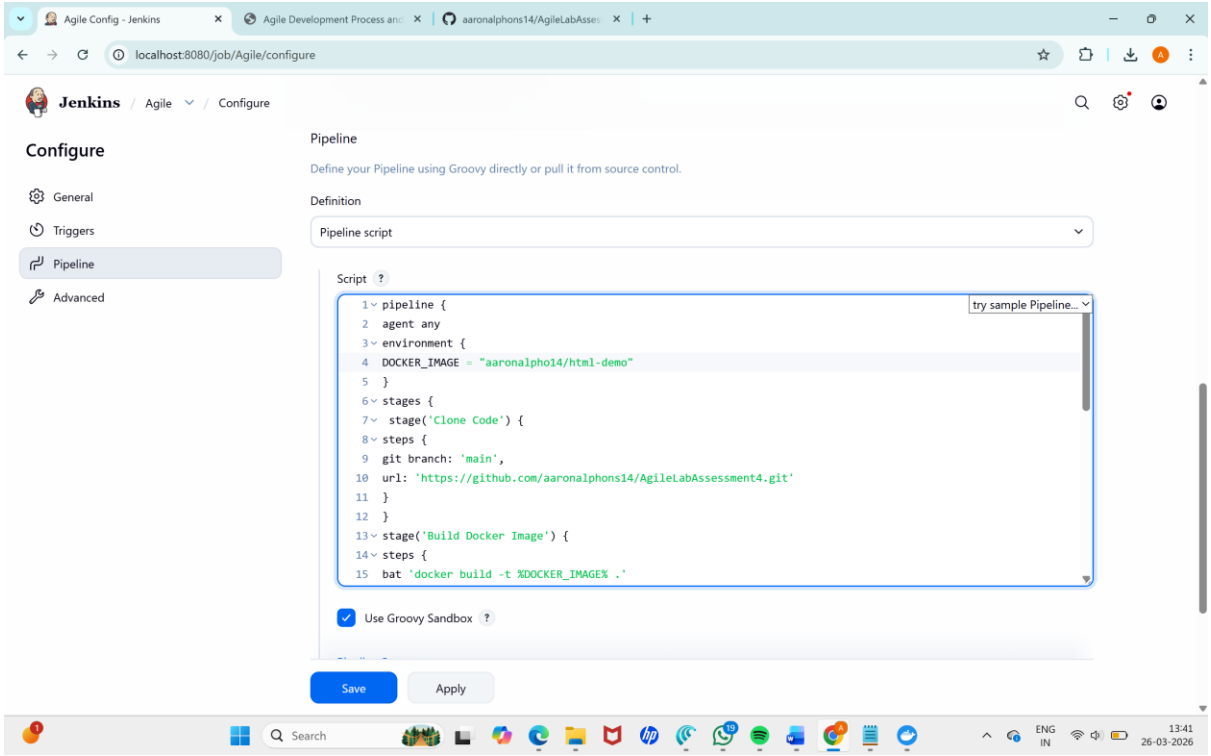
3. Open Jenkins In Jenkins Setting --> Credentials New Credentials – User name: , password , ID:, Description

New Credentials – Secret – Broswse C: User->Administrator-->.kube





Create New Item -> Pipeline Paste the code in pipeline script - Update URL of GIT repository - Make sure credentialsId: “dockerhub” same as mentioned in credentials - Make sure that the ID mentioned in credentials “kuberconfig” is same as in script - Run the Build Pipeline:



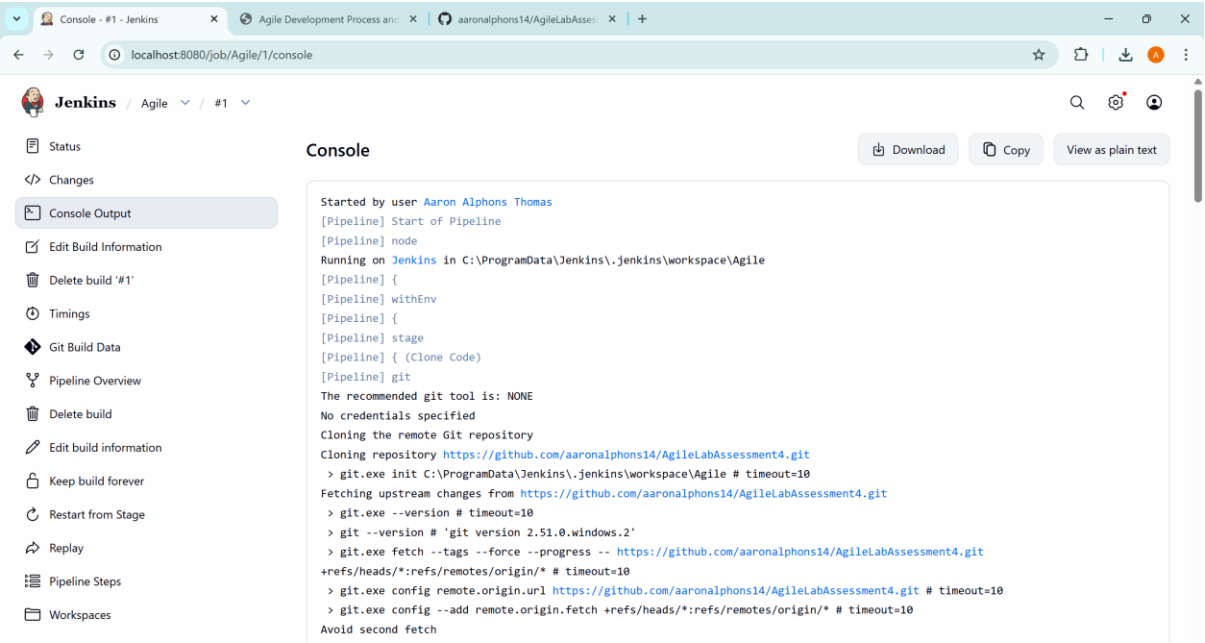
Docker Pipeline

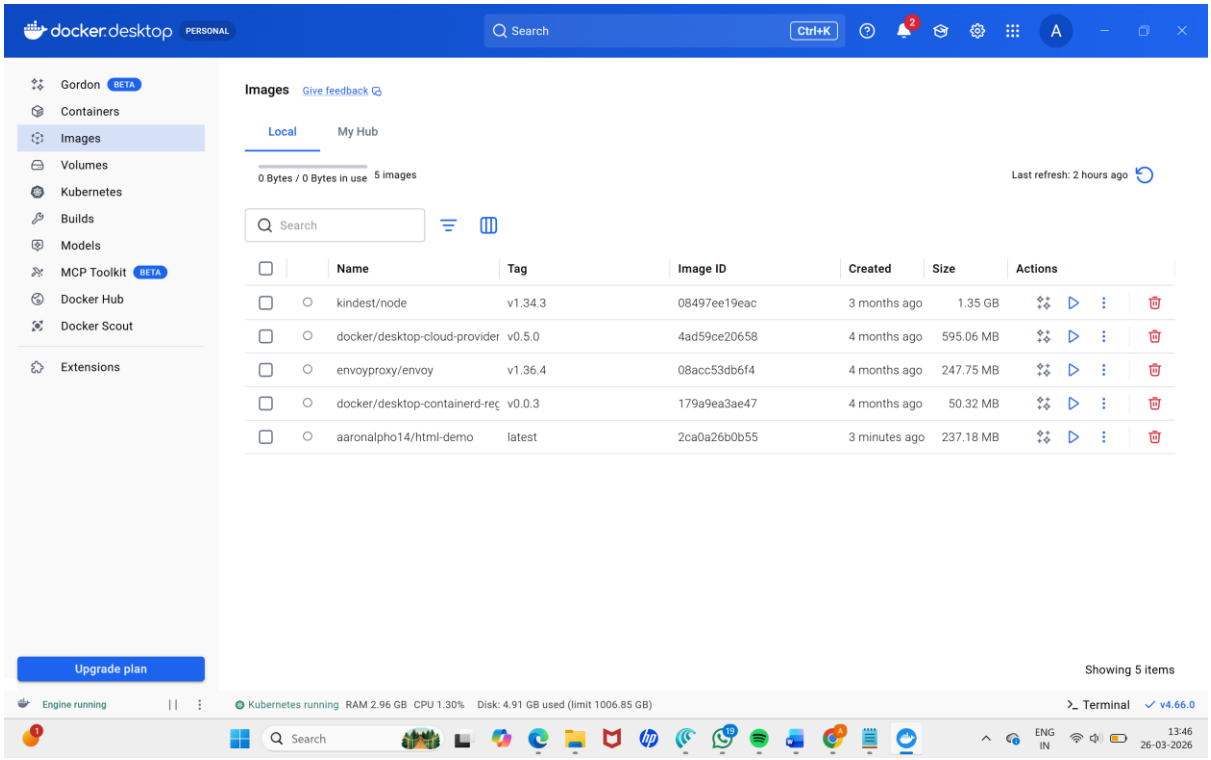
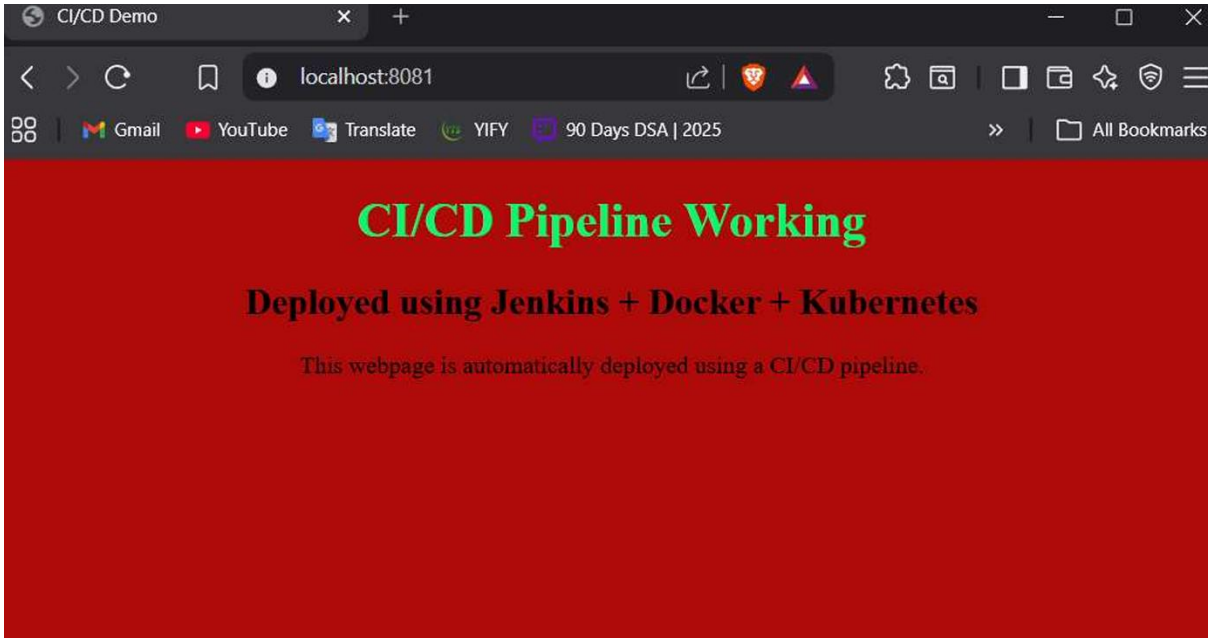
Stage View

	Clone Code	Build Docker Image	Push Image	Deploy to Kubernetes
Average stage times: (full run time: ~14s)	1s	1s	9s	812ms
#5 Mar 25 18:35 No Changes	1s	1s	9s	812ms

Permalinks

- Last build (#5), 12 min ago
- Last stable build (#5), 12 min ago
- Last successful build (#5), 12 min ago
- Last completed build (#5), 12 min ago


Console - #1 - Jenkins
Agile Development Process and
aaronalphons14/AgileLabAsses
localhost:8080/job/Agile/1/console
Jenkins / Agile / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Git Build Data
Pipeline Overview
Delete build
Edit build information
Keep build forever
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as plain text
Started by user Aaron Alphons Thomas
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\Agile
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Code)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/aaronalphons14/AgileLabAssessment4.git
> git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\Agile # timeout=10
Fetching upstream changes from https://github.com/aaronalphons14/AgileLabAssessment4.git
> git.exe --version # timeout=10
> git --version # 'git version 2.51.0.windows.2'
> git.exe fetch --tags --force --progress -- https://github.com/aaronalphons14/AgileLabAssessment4.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe config remote.origin.url https://github.com/aaronalphons14/AgileLabAssessment4.git # timeout=10
> git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch


docker desktop PERSONAL
Search Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Images Give feedback
Local
My Hub
0 Bytes / 0 Bytes in use 5 images
Last refresh: 2 hours ago
Search
Name Tag Image ID Created Size Actions
kindest/node v1.34.3 08497ee19eac 3 months ago 1.35 GB
docker/desktop-cloud-provider v0.5.0 4ad59ce20658 4 months ago 595.06 MB
envoyproxy/envoy v1.36.4 08acc53db6f4 4 months ago 247.75 MB
docker/desktop-containerd-reg v0.0.3 179a9ea3ae47 4 months ago 50.32 MB
aaronalpho14/html-demo latest 2ca0a26b0b55 3 minutes ago 237.18 MB
Showing 5 items
Engine running
Kubernetes running RAM 2.96 GB CPU 1.30% Disk: 4.91 GB used (limit 1006.85 GB)
Terminal
v4.66.0
Search
ENG IN
13:46 26-03-2026

nodePort:8081 (after redirecting from 30010)